Seminar 3 - Lectures 5, 6 and 7
Computer Vision 1, Master AI, 2026

Exercises inherited from Arnoud Visser, Martin Oswald, Qi Bi, Roan van Blanken

1 Exercise 1: Edges and Corners

In this exercise, we explore key structures in images: edges and corners. At these interest points, there are sharp differences in brightness, which can indicate various phenomena, such as changes in depth, material, and illumination. Edges and corners are fundamental features in classical computer vision and are widely used in more complex image processing algorithms.

We consider two simple image patches, A and B, from a random 4-bit image shown below. The patches are highlighted in yellow. The values in the cells represent the intensity values in the range $[0, 15]$ at different locations in the image.

$A =$

0	0	6	13	12
0	0	6	13	12
0	0	6	13	12
0	0	6	13	12
0	0	6	13	12

$B =$

0	0	0	0	0
0	0	0	0	0
12	7	7	0	0
12	12	7	0	0
12	12	10	0	0

Q.1.a What structures do you see in image patches A and B?

For the next couple of questions, we use the following partial derivatives: $I_x = \frac{\partial I}{\partial x}$ and $I_y = \frac{\partial I}{\partial y}$. To compute these, we apply the following simple correlation kernels. These kernels are already flipped, and we omit the normalization constant $\frac{1}{2}$:

Correlation kernel (x-direction):

-1	0	1

Correlation kernel (y-direction):

-1
0
1

Q.1.b Cross-correlate patch A with the derivative kernel. Compute the patch derivatives in x and y directions. Compute the gradient magnitude M_A.

Optional

The Harris Transform is used to estimate the *cornerness* H of an image patch. This measure helps in detecting corners, edges, and flat regions by analyzing the intensity variations within the patch. A simplified definition of the *cornerness* is given by:

$$H = \lambda_1 \lambda_2 - 0.04(\lambda_1 + \lambda_2)^2$$

where λ_1 and λ_2 are the eigenvalues of the structure tensor matrix M. An alternative formulation, which avoids computing eigenvalues directly, is given by:

$$H = \det(M) - 0.04 \cdot (\text{trace}(M))^2$$

where:

- $\det(M)$ is the determinant of matrix M,
- $\text{trace}(M)$ is the sum of the diagonal elements of matrix M (i.e., $M_{11} + M_{22}$).

Matrix M, known as the structure tensor, encapsulates information about the gradient of the image patch and is defined as:

$$M = \sum_{(x,y) \in W} \begin{bmatrix} I_x(x,y)^2 & I_x(x,y)I_y(x,y) \\ I_x(x,y)I_y(x,y) & I_y(x,y)^2 \end{bmatrix}$$

Here:

- $I_x(x,y)$ and $I_y(x,y)$ represent the partial derivatives of the image intensity with respect to x and y at pixel location (x,y),
- W is the image patch (a window of pixels) over which the structure tensor M is computed.

In essence, the Harris Transform evaluates the cornerness of a patch by analyzing how the intensity gradients I_x and I_y vary within the patch. The matrix M captures this information, and the value of H helps classify the patch as an edge, a flat region, or a corner, depending on the eigenvalues of M.

Q.1.c Compute the Harris corner response for patch A. What can you conclude based on your result?

Q.1.d Now repeat the exercise for patch B. Compute the gradient magnitude and the Harris corner response of image patch B. What can you conclude based on your results?

Q.1.e Alternatively, we could use the eigenvalues of structure tensor M. These eigenvalues can also help us make a statement about the *cornerness* of the patch. Compute the eigenvalues of the structure tensor M for patch B. What do these values indicate?

2 Exercise 2: Histogram of Gradients

Consider the following image patch (I):

$$I_{\mathrm{A}} = \begin{array}{|c|c|c|c|c|} \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 0 \\ \hline 1 & 1 & 1 & 0 & 0 \\ \hline 1 & 1 & 0 & 0 & 0 \\ \hline 1 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

Q.2.a Compute the gradient G_x and G_y, using image filters. Which filters do you use?

Q.2.b Compute the gradient magnitude

Q.2.c Compute the gradient orientation (in degrees)

Q.2.d Compute the HoG descriptor, using a 9-bin histogram

Q.2.e Is the HoG descriptor invariant to overall lighting, *i.e.*, is the HoG descriptor of $I = 2 * I_A$ equal to the HoG descriptor of I_A?

Q.2.f Is the HoG descriptor invariant to rotation, *i.e.*, is the HoG descriptor of I_A equal to the HoG descriptor of I_A after a 90-degree rotation?

3 Exercise 3: SIFT

The SIFT (Scale Invariant Feature Transform) has become a popular technique for detecting and representing features for structure-from-motion and object recognition tasks.

Q.3.a Explain how feature locations are determined in the SIFT operator.

Q.3.b Explain how the SIFT operator achieves invariance to (i) illumination; (ii) scale; (iii) rotation.

Q.3.c A SIFT descriptor for a 16×16 image window $\mathbf{W}$ is a vector of 128 numbers, which can be thought of as being grouped in 16 consecutive groups of 8 numbers each. What does each group of 8 numbers represent? Explain in one or two brief, clear, and accurate sentences.

4 Exercise 4: Optical Flow

Q.4.a For the Lukas-Kanade optical flow technique, mention two of the main assumptions

Q.4.b How can we use optical flow for traffic monitoring i.e. ticketing higher than allowed speeds?

Q.4.c How could you use optical flow to add effects like slow motion or video smoothing and when this doesn't work?

Q.4.d How can we use optical flow for shaking camera and image stabilization?

Q.4.e How we can use optical flows to estimate depth?

Q.4.f What is the difference between dense and sparse optical flows? What is the pros and cons of each?

5 Exercise 5: 2D Image Warping

Q.5.a For 2D transformations, write down an example transformation matrix for a rotation for homogeneous coordinates.

Q.5.b For the same dimensions, give an example *affine* transformation.

Q.5.c Explain difference between forward and backward transformation

6 Exercise 6: 3D Global Transformation

Write the homogeneous 4x4 matrices for the following transforms:

Q.6.a Translate by +5 units in the X direction

Q.6.b Rotate by 30 degrees about the X axis

Q.6.c The rotation, followed by the translation above, followed by scaling by a factor of 2.